

# Reproducing the experiments of Derrode and Pieczynski (CSDA, 2013) with the `awesomePMC` package

Reproducibility report — general PMC with pair-indexed margins

14 September 2026

## Abstract

We reproduce the simulation experiments of sections 3.2, 3.3 and 4.3 of Derrode and Pieczynski [1] with the open-source `awesomePMC` package. Three findings shape the reproduction. *(i)* The model of the paper is the general pairwise Markov chain whose margins  $f_{ij}$  are indexed by the *pair* of states (Eqs. 12–14 of the paper); earlier versions of the package collapsed them to one margin per state, which by the Proposition of the paper’s §2.1 turns the PMC into a hidden Markov chain and put the diagonal error rates 7 to 19 points above the paper’s on average. *(ii)* The second parameter of the Gaussian margins of the paper’s Table 1 is a variance: only that reading makes the printed Gamma parameters consistent. *(iii)* The unsupervised experiment needs several ICE starts: with Gaussian margins, a single start ends on a copula layout of markedly lower likelihood. With the general PMC, Tables 2(a), 2(b), 3(b) and 4(a) are reproduced to within half a point on average over their 49 cells; with a multistart ICE, the diagonal copulas of Tables 6–7 are recovered in 97–100 % of runs and the unsupervised error comes within half a point of the supervised one. We document the gaps that remain: Table 3(a) (Gaussian margins,  $\tau = 0.70$ ) and the PMM Tables 4(b) and 5(b).

## 1 Introduction

The CSDA-2013 paper introduces an unsupervised classification method for sequences modelled by a *stationary and reversible pairwise Markov chain* (SR-PMC) whose bivariate observation densities are built from copulas. Three sets of experiments illustrate, in order:

1. the influence of the copula *shape* on supervised restoration, when the simulation copula and the restoration copula vary independently (paper §3.2, Tables 2–3; our Section 4);
2. the same study with the i.i.d. *pairwise mixture model* (PMM), which isolates the effect of Markovianity (paper §3.3, Tables 4–5; Section 5);
3. the recovery of the copula families and parameters by ICE with automatic copula selection (paper §4.3, Tables 6–7; Section 7).

Section 8 adds a comparison of six selection criteria that is not in the paper. The radar-image segmentation of the paper’s §5 is not reproduced.

This report replaces an earlier version whose numbers were produced with one margin per state and a single ICE start. Section 2 explains why both choices were wrong and how the reproduction was corrected; the earlier results are kept in the repository and are quoted where the comparison is informative.

## 2 Setup and notation

### 2.1 The general PMC with copulas

A PMC is a Markov chain  $Z_n = (X_n, Y_n)$  with hidden states  $X_n \in \{1, \dots, K\}$  and observations  $Y_n$ . In the stationary and reversible case considered by the paper, its law is given by the joint probabilities

$p_{ij} = p(x_n = i, x_{n+1} = j)$ ,  $p_{ij} = p_{ji}$ , and by the density of a pair of consecutive observations given the pair of states (paper, Eq. 12):

$$f(y_n, y_{n+1} \mid x_n = i, x_{n+1} = j) = f_{ij}(y_n) f_{ji}(y_{n+1}) c_{ij}(F_{ij}(y_n), F_{ji}(y_{n+1})), \quad (1)$$

where  $f_{ij}$  is the law of  $y_n$  given  $(x_n, x_{n+1}) = (i, j)$ ,  $F_{ij}$  its CDF and  $c_{ij}$  a copula density. Reversibility only makes the right margin of the pair  $(i, j)$  the left margin of the pair  $(j, i)$  — the index inversion  $f_{ji}$  in Eq. (1). The transitions follow (paper, Eqs. 13–14):

$$p(x_{n+1} = j \mid x_n = i, y_n) \propto p_{ij} f_{ij}(y_n), \quad (2)$$

$$p(y_{n+1} \mid x_n = i, x_{n+1} = j, y_n) = f_{ji}(y_{n+1}) c_{ij}(F_{ij}(y_n), F_{ji}(y_{n+1})). \quad (3)$$

**State margins make  $X$  Markov.** The Proposition of the paper’s §2.1 states that, for an SR-PMC,  $X$  is a Markov chain if and only if  $p(y_2 \mid x_1, x_2) = p(y_2 \mid x_2)$ , i.e. if and only if  $f_{ij} = f_i$  does not depend on  $j$ . Versions 0.5.0 to 0.8 of the package imposed  $f_{ij} = f_i$ , attributing it to reversibility. The model called “PMC” was then a stationary reversible HMC with correlated noise, not the model of the paper. The package now keeps both structures, selected by `[model].margin_structure`: “state” ( $K$  margins  $f_i$ ) and “pair” ( $K^2$  margins  $f_{ij}$ , the general PMC).

## 2.2 Margins of the paper’s Table 1

Table 1 of the paper gives four margins for  $K = 2$ : Gaussian  $\mathcal{N}(\mu, s)$  with  $(\mu, s) = (0, 1), (0.3, 1.6), (1.1, 1.4), (1.5, 1)$  for  $f_{11}, f_{12}, f_{21}, f_{22}$ , and Gamma margins  $\mathcal{G}(\lambda, \alpha, \theta)$  “chosen so that their standard deviations are identical to those of Gaussian distributions”.

**The second Gaussian parameter is a variance.** The printed Gamma parameters are consistent with the caption only if  $\lambda$  is the location of the *centred* Gamma variable (the mean  $\mu$  being added afterwards),  $\theta$  is the square of the scale, and  $s$  is a *variance*. For a Gamma margin of shape  $\alpha$  and standard deviation  $\sigma$  these quantities are  $\lambda = -\sqrt{\alpha} \sigma$  and  $\theta = \sigma^2/\alpha$ . Table 1 compares the printed values with the two readings of  $s$ : reading  $s$  as a variance reproduces  $f_{11}, f_{21}$  and  $f_{22}$  up to the rounding of the printed values and  $f_{12}$  to 0.12 on  $\lambda$ , whereas reading it as a standard deviation misses  $f_{12}$  and  $f_{21}$  entirely.

Table 1: Gamma parameters of Table 1: printed, and implied by each reading of the Gaussian parameter  $s$  ( $\lambda = -\sqrt{\alpha} \sigma$ ,  $\theta = \sigma^2/\alpha$ , with  $\sigma = s$  or  $\sigma = \sqrt{s}$ ).

| margin   | Gaussian $\mathcal{N}(\mu, s)$ |      | printed $\mathcal{G}$ |          | $s = \text{std}$ |          | $s = \text{variance}$ |          |
|----------|--------------------------------|------|-----------------------|----------|------------------|----------|-----------------------|----------|
|          | $\mu$                          | $s$  | $\lambda$             | $\theta$ | $\lambda$        | $\theta$ | $\lambda$             | $\theta$ |
| $f_{11}$ | 0.0                            | 1.00 | -2.83                 | 0.13     | -2.83            | 0.125    | -2.83                 | 0.125    |
| $f_{12}$ | 0.3                            | 1.60 | -3.65                 | 0.18     | -4.77            | 0.288    | -3.77                 | 0.180    |
| $f_{21}$ | 1.1                            | 1.40 | -2.08                 | 0.46     | -2.46            | 0.636    | -2.08                 | 0.455    |
| $f_{22}$ | 1.5                            | 1.00 | -1.63                 | 0.38     | -1.63            | 0.375    | -1.63                 | 0.375    |

The supervised restorations agree: with pair margins, reading  $s$  as a variance brings the diagonal cells of Tables 2(a), 2(b) and 3(b) within 0.5 point of the paper on average, against about 1.2 points with the standard deviation (30 replications per cell). All results below use pair margins with  $s$  read as a variance; the Gamma margins use the printed shapes  $\alpha$  with location and scale matched to  $(\mu, \sqrt{s})$ .

## 2.3 Reading the paper’s tables

In Tables 2–5 of the paper, rows are the *tested* (restoration) copula and columns the *true* (simulation) copula — “when the true copula is  $c^3$  and the tested copula is  $c^5$ , we get an error of 14.11 %” — and the minimum of each column, i.e. the best restoration copula for each true copula, is highlighted. Our tables follow the same orientation. (The earlier version of this report printed the transpose.)

## 2.4 Copulas under test

The copula codes  $c^0, \dots, c^8$  are recalled in Appendix A. The eligible sets of the supervised experiments contain the copulas that admit the target Kendall’s  $\tau$  (Table 2).

Table 2: Eligible-copula sets for the two correlation regimes.

| Regime              | Set                                                 |
|---------------------|-----------------------------------------------------|
| $\tau^{(1)} = 0.16$ | $\Pi^{(1)} = \{c^0, c^1, c^2, c^3, c^4, c^5, c^6\}$ |
| $\tau^{(2)} = 0.70$ | $\Pi^{(2)} = \{c^0, c^1, c^2, c^3, c^6, c^7, c^8\}$ |

## 3 Reproducibility

The results come from the script `report/reproduce_csda2013.py` and every table from `report/csda2013_tables.py`, which reads the CSV results and the values of the paper (`report/data/csda2013_tables2to5.csv`). The diagnostics of Sections 2.2 and 7.2 come from the scripts kept with the working notes (`report/out/csda_compare/`, not versioned). The runs reported here are:

```
# Experiments 1-2 (Tables 2-5): 30 replications per cell, N = 2000
python report/reproduce_csda2013.py --exp1 --exp2 --reps 30 --margins pair
# Experiment 3 (Tables 6-7), single ICE start: 100 runs
python report/reproduce_csda2013.py --exp3 --runs 100 --margins pair
# Experiment 3, multistart ICE: 30 runs
python report/reproduce_csda2013.py --exp3 --runs 30 --margins pair --exp3-starts multi
# Tables of this report
python report/csda2013_tables.py
```

`-margins pair` reads  $s$  as a variance by default (`-table1-sigma`); `-margins state` reproduces the earlier one-margin-per-state results. The paper used 300 replications per cell and 10 ICE runs. Every experiment uses explicit seeds. With five worker processes on a laptop, Experiments 1–2 take a few minutes, the single-start Experiment 3 about 40 minutes and the multistart one about 3.5 hours.

## 4 Experiment 1 — Supervised PMC restoration (CSDA Tables 2 and 3)

### 4.1 Setup

$K = 2$ ,  $N = 2000$  observations per sequence, and the prior  $p(1, 1) = 0.50$ ,  $p(1, 2) = p(2, 1) = 0.05$ ,  $p(2, 2) = 0.40$ . For each copula of  $\Pi^{(\bullet)}$  we simulate  $(X_{1:N}, Y_{1:N})$  with that copula on all four transitions, restore  $X$  by the MPM classifier with each copula of the same set (again on all four transitions, every other parameter being the true one), and record the error rate. Each cell averages 30 replications; the empirical standard deviation is given in parentheses.

### 4.2 Results

### 4.3 Discussion

The qualitative findings of the paper are recovered:

- For every true copula, the best restoration copula is the true one (27 columns out of 28; in the last one,  $c^4$  at  $\tau = 0.16$  with Gaussian margins,  $c^1$  does better by 0.01 point).
- At  $\tau = 0.16$  the copulas are nearly interchangeable: a wrong choice raises the error rate by at most 10 % of its value.

Table 3: PMC supervised —  $\tau = 0.16$ , Gaussian margins (CSDA Table 2(a)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $\mathbf{c}^0$     | $\mathbf{c}^1$     | $\mathbf{c}^2$     | $\mathbf{c}^3$     | $\mathbf{c}^4$     | $\mathbf{c}^5$     | $\mathbf{c}^6$     |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>10.76</b> (0.9) | 13.85 (1.1)        | 13.40 (1.3)        | 13.66 (1.1)        | 14.27 (1.0)        | 14.18 (1.2)        | 13.74 (1.1)        |
| $c^1$         | 10.82 (0.9)        | <b>13.66</b> (1.1) | 13.35 (1.4)        | 13.60 (1.1)        | <b>14.13</b> (1.0) | 13.90 (1.1)        | 13.58 (1.2)        |
| $c^2$         | 11.33 (1.0)        | 14.12 (1.1)        | <b>12.94</b> (1.2) | 13.82 (1.1)        | 14.83 (1.0)        | 14.87 (1.2)        | 14.04 (1.2)        |
| $c^3$         | 10.97 (1.0)        | 13.85 (1.1)        | 13.29 (1.2)        | <b>13.52</b> (1.1) | 14.46 (1.0)        | 14.28 (1.0)        | 14.18 (1.2)        |
| $c^4$         | 10.97 (0.9)        | 13.72 (1.1)        | 13.49 (1.4)        | 13.76 (1.1)        | 14.14 (1.0)        | 13.84 (1.2)        | 13.63 (1.2)        |
| $c^5$         | 11.22 (1.0)        | 13.83 (1.1)        | 13.94 (1.4)        | 13.89 (1.1)        | 14.22 (1.1)        | <b>13.81</b> (1.2) | 13.68 (1.2)        |
| $c^6$         | 11.28 (0.9)        | 13.93 (1.2)        | 13.56 (1.2)        | 14.15 (1.2)        | 14.43 (1.0)        | 14.26 (1.2)        | <b>13.18</b> (1.2) |

Table 4: PMC supervised —  $\tau = 0.16$ , Gamma margins (CSDA Table 2(b)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $\mathbf{c}^0$    | $\mathbf{c}^1$     | $\mathbf{c}^2$     | $\mathbf{c}^3$     | $\mathbf{c}^4$     | $\mathbf{c}^5$    | $\mathbf{c}^6$    |
|---------------|-------------------|--------------------|--------------------|--------------------|--------------------|-------------------|-------------------|
| $c^0$         | <b>8.01</b> (0.7) | 10.63 (0.8)        | 10.64 (1.1)        | 10.87 (1.1)        | 10.54 (1.1)        | 10.37 (0.9)       | 10.20 (0.9)       |
| $c^1$         | 8.50 (0.8)        | <b>10.02</b> (0.7) | 10.50 (1.3)        | 10.54 (1.1)        | 10.12 (1.0)        | 9.98 (0.9)        | 9.54 (0.9)        |
| $c^2$         | 8.63 (0.8)        | 10.47 (0.8)        | <b>10.00</b> (1.2) | 10.80 (1.1)        | 10.79 (1.1)        | 10.64 (1.0)       | 9.75 (0.8)        |
| $c^3$         | 8.29 (0.8)        | 10.18 (0.7)        | 10.35 (1.3)        | <b>10.43</b> (1.0) | 10.40 (1.1)        | 10.20 (0.9)       | 9.88 (0.9)        |
| $c^4$         | 8.42 (0.7)        | 10.12 (0.7)        | 10.63 (1.1)        | 10.59 (1.1)        | <b>10.02</b> (1.0) | 9.89 (1.0)        | 9.55 (0.9)        |
| $c^5$         | 8.55 (0.7)        | 10.21 (0.7)        | 10.88 (1.2)        | 10.68 (1.1)        | 10.07 (1.1)        | <b>9.85</b> (0.9) | 9.65 (1.0)        |
| $c^6$         | 8.77 (0.8)        | 10.29 (0.7)        | 10.38 (1.1)        | 10.97 (1.1)        | 10.37 (1.0)        | 10.23 (1.0)       | <b>9.38</b> (0.9) |

Table 5: PMC supervised —  $\tau = 0.70$ , Gaussian margins (CSDA Table 3(a)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $\mathbf{c}^0$     | $\mathbf{c}^1$     | $\mathbf{c}^2$     | $\mathbf{c}^3$     | $\mathbf{c}^6$     | $\mathbf{c}^7$     | $\mathbf{c}^8$     |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>10.76</b> (0.9) | 25.22 (2.5)        | 24.08 (2.1)        | 24.69 (1.9)        | 27.18 (2.1)        | 24.94 (1.9)        | 24.56 (2.5)        |
| $c^1$         | 41.40 (0.9)        | <b>20.68</b> (1.8) | 23.21 (2.5)        | 23.47 (1.9)        | 25.76 (2.2)        | 22.98 (2.4)        | 22.34 (2.8)        |
| $c^2$         | 30.25 (0.9)        | 21.96 (1.9)        | <b>22.45</b> (2.4) | 22.78 (1.7)        | 24.84 (2.2)        | 22.32 (2.1)        | 22.00 (2.6)        |
| $c^3$         | 40.61 (1.7)        | 24.49 (2.2)        | 24.88 (2.7)        | <b>18.07</b> (1.9) | 36.00 (2.3)        | 26.69 (2.4)        | 21.50 (2.5)        |
| $c^6$         | 46.73 (1.8)        | 26.22 (1.7)        | 27.11 (2.6)        | 33.73 (1.9)        | <b>10.42</b> (1.4) | 23.78 (2.4)        | 29.64 (2.7)        |
| $c^7$         | 35.72 (1.1)        | 22.48 (2.0)        | 23.47 (2.5)        | 27.35 (1.9)        | 17.52 (1.7)        | <b>21.42</b> (2.0) | 23.80 (2.8)        |
| $c^8$         | 36.30 (1.1)        | 22.50 (1.9)        | 23.53 (2.6)        | 19.37 (1.8)        | 31.84 (2.3)        | 24.22 (2.2)        | <b>20.76</b> (2.6) |

Table 6: PMC supervised —  $\tau = 0.70$ , Gamma margins (CSDA Table 3(b)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $\mathbf{c}^0$    | $\mathbf{c}^1$     | $\mathbf{c}^2$     | $\mathbf{c}^3$     | $\mathbf{c}^6$    | $\mathbf{c}^7$     | $\mathbf{c}^8$     |
|---------------|-------------------|--------------------|--------------------|--------------------|-------------------|--------------------|--------------------|
| $c^0$         | <b>8.01</b> (0.7) | 22.81 (2.4)        | 22.64 (2.6)        | 22.36 (2.4)        | 23.72 (2.2)       | 21.85 (2.3)        | 22.69 (2.1)        |
| $c^1$         | 45.12 (2.1)       | <b>13.78</b> (2.0) | 16.37 (2.2)        | 19.20 (2.4)        | 12.24 (1.9)       | 14.07 (2.3)        | 17.34 (2.0)        |
| $c^2$         | 30.94 (1.8)       | 14.92 (2.1)        | <b>15.27</b> (2.0) | 19.77 (2.2)        | 9.35 (1.6)        | 13.19 (2.2)        | 17.19 (2.0)        |
| $c^3$         | 35.72 (1.4)       | 16.26 (2.1)        | 17.35 (2.1)        | <b>17.84</b> (2.2) | 17.49 (2.3)       | 15.98 (2.4)        | 17.68 (2.3)        |
| $c^6$         | 46.02 (2.4)       | 18.05 (2.1)        | 17.58 (2.1)        | 23.27 (2.3)        | <b>6.85</b> (1.2) | 14.64 (2.3)        | 20.24 (2.1)        |
| $c^7$         | 37.90 (1.9)       | 14.98 (2.2)        | 15.50 (1.9)        | 20.23 (2.3)        | 8.61 (1.5)        | <b>13.04</b> (2.3) | 17.41 (2.0)        |
| $c^8$         | 35.78 (1.5)       | 14.71 (2.1)        | 16.01 (2.0)        | 18.52 (2.2)        | 12.58 (1.7)       | 13.81 (2.4)        | <b>16.80</b> (2.0) |

- At  $\tau = 0.70$  the choice matters: restoring Clayton data ( $c^6$ ) with a Gumbel copula ( $c^3$ ) raises the error from 10.4 % to 36.0 % with Gaussian margins, and restoring independent data with a Clayton copula multiplies it by 4.3 (Gaussian) and 5.7 (Gamma).

The agreement with the paper’s numbers is quantified in Section 6.

## 5 Experiment 2 — Pairwise mixture model (CSDA Tables 4 and 5)

### 5.1 Setup

In the PMM, the pairs  $(y_1, y_2), (y_3, y_4), \dots$  are independent. We draw  $N/2 = 1000$  pairs, the first observation from  $f_{ij}$  and the second from  $f_{ji}$  with copula  $c_{ij}$ , and classify each pair with the posterior

$$p(x_1 = i \mid y_1, y_2) \propto \sum_{j=1}^K p_{ij} f_{ij}(y_1) f_{ji}(y_2) c_{ij}(F_{ij}(y_1), F_{ji}(y_2)). \quad (4)$$

The (simulation, restoration) combinations are those of Experiment 1.

### 5.2 Results

Table 7: PMM (i.i.d.) supervised —  $\tau = 0.16$ , Gaussian margins (CSDA Table 4(a)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $c^0$              | $c^1$              | $c^2$              | $c^3$              | $c^4$              | $c^5$              | $c^6$              |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>17.95</b> (0.9) | 20.09 (1.1)        | 19.15 (1.0)        | 19.84 (0.9)        | 20.85 (1.4)        | 20.87 (1.1)        | 19.57 (1.1)        |
| $c^1$         | 17.97 (0.9)        | <b>20.06</b> (1.0) | 19.31 (1.0)        | 19.89 (0.9)        | 20.79 (1.4)        | <b>20.71</b> (1.1) | 19.52 (1.0)        |
| $c^2$         | 17.98 (0.9)        | 20.17 (1.1)        | <b>19.11</b> (1.0) | 19.85 (0.9)        | 20.89 (1.5)        | 20.85 (1.1)        | 19.61 (1.0)        |
| $c^3$         | 18.05 (0.8)        | 20.19 (1.1)        | 19.22 (1.0)        | <b>19.75</b> (0.9) | 20.83 (1.3)        | 20.77 (1.1)        | 19.80 (1.1)        |
| $c^4$         | 17.99 (0.9)        | 20.13 (1.0)        | 19.36 (1.0)        | 19.94 (0.9)        | 20.78 (1.4)        | 20.71 (1.0)        | 19.49 (1.0)        |
| $c^5$         | 18.23 (1.0)        | 20.15 (1.0)        | 19.54 (1.0)        | 20.07 (0.9)        | <b>20.73</b> (1.4) | 20.71 (1.0)        | 19.55 (1.0)        |
| $c^6$         | 18.12 (1.1)        | 20.20 (1.0)        | 19.46 (1.1)        | 20.18 (0.9)        | 20.88 (1.4)        | 20.74 (1.0)        | <b>19.39</b> (1.1) |

Table 8: PMM (i.i.d.) supervised —  $\tau = 0.16$ , Gamma margins (CSDA Table 4(b)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $c^0$              | $c^1$              | $c^2$              | $c^3$              | $c^4$              | $c^5$              | $c^6$              |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>13.74</b> (0.9) | 16.42 (1.0)        | 16.22 (0.9)        | 16.43 (1.2)        | 16.75 (1.2)        | 16.73 (1.0)        | 16.35 (1.0)        |
| $c^1$         | 13.96 (0.9)        | 16.32 (1.0)        | 16.27 (0.9)        | 16.44 (1.2)        | <b>16.60</b> (1.2) | 16.45 (1.0)        | 16.02 (1.0)        |
| $c^2$         | 13.90 (0.8)        | 16.40 (1.0)        | <b>15.95</b> (0.8) | 16.53 (1.2)        | 16.68 (1.2)        | 16.70 (0.9)        | 16.12 (1.0)        |
| $c^3$         | 13.75 (0.8)        | 16.39 (1.0)        | 16.20 (0.9)        | <b>16.41</b> (1.2) | 16.72 (1.2)        | 16.59 (1.0)        | 16.20 (1.0)        |
| $c^4$         | 13.94 (0.9)        | <b>16.31</b> (1.0) | 16.25 (0.9)        | 16.44 (1.2)        | 16.63 (1.2)        | 16.47 (1.0)        | 16.01 (1.0)        |
| $c^5$         | 14.09 (0.9)        | 16.34 (1.0)        | 16.48 (1.0)        | 16.55 (1.2)        | 16.64 (1.2)        | <b>16.44</b> (1.0) | 16.09 (1.1)        |
| $c^6$         | 14.37 (0.9)        | 16.40 (0.9)        | 16.30 (0.8)        | 16.75 (1.3)        | 16.71 (1.1)        | 16.58 (1.0)        | <b>15.97</b> (1.0) |

### 5.3 Discussion

Markovianity lowers the error of the correct restoration (diagonal cells) in every configuration. At  $\tau = 0.16$  the PMC error is 32–40 % lower than the PMM one with Gaussian margins and 36–42 % lower with Gamma margins, in line with the paper. At  $\tau = 0.70$  the gain depends on the copula: with Gaussian margins it is 40–48 % for the product and Clayton copulas but only 6–21 % for  $c^1$ ,  $c^2$ ,  $c^3$ ,  $c^7$  and  $c^8$ ; with Gamma margins it is 17–49 %. The small gains with Gaussian margins at  $\tau = 0.70$  come from the PMC side, where our errors exceed the paper’s (Table 3(a), Section 6).

Table 9: PMM (i.i.d.) supervised —  $\tau = 0.70$ , Gaussian margins (CSDA Table 5(a)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $c^0$              | $c^1$              | $c^2$              | $c^3$              | $c^6$              | $c^7$              | $c^8$              |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>17.95</b> (0.9) | 24.74 (1.4)        | 24.62 (1.2)        | 24.68 (1.0)        | 25.56 (1.3)        | 24.05 (1.0)        | 24.59 (1.1)        |
| $c^1$         | 30.85 (0.9)        | <b>23.25</b> (1.3) | 23.92 (1.1)        | 23.86 (0.9)        | 24.18 (1.3)        | 23.34 (1.1)        | 23.73 (1.0)        |
| $c^2$         | 26.40 (0.8)        | 23.49 (1.4)        | <b>23.89</b> (1.1) | 23.91 (0.9)        | 24.02 (1.2)        | 23.30 (1.1)        | 23.77 (1.0)        |
| $c^3$         | 30.46 (0.9)        | 24.29 (1.4)        | 24.54 (1.3)        | <b>23.00</b> (1.1) | 27.78 (1.2)        | 24.52 (1.1)        | 23.73 (1.2)        |
| $c^6$         | 36.02 (1.2)        | 26.58 (1.0)        | 26.78 (1.2)        | 29.70 (1.3)        | <b>20.16</b> (1.0) | 25.67 (1.1)        | 28.01 (1.2)        |
| $c^7$         | 29.06 (0.9)        | 23.65 (1.3)        | 23.98 (1.1)        | 24.66 (1.0)        | 23.27 (1.3)        | <b>23.14</b> (1.1) | 24.05 (1.0)        |
| $c^8$         | 29.46 (0.8)        | 23.69 (1.4)        | 24.12 (1.2)        | 23.36 (1.1)        | 25.83 (1.1)        | 23.60 (1.1)        | <b>23.46</b> (1.0) |

Table 10: PMM (i.i.d.) supervised —  $\tau = 0.70$ , Gamma margins (CSDA Table 5(b)). Pair-indexed margins  $f_{ij}$  of Table 1, second Gaussian parameter read as a variance.

| tested \ true | $c^0$              | $c^1$              | $c^2$              | $c^3$              | $c^6$              | $c^7$              | $c^8$              |
|---------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $c^0$         | <b>13.74</b> (0.9) | 22.89 (1.1)        | 22.41 (1.1)        | 22.54 (1.3)        | 23.12 (1.2)        | 22.50 (1.1)        | 23.10 (1.3)        |
| $c^1$         | 31.05 (0.9)        | <b>19.55</b> (1.0) | 20.21 (1.1)        | 22.17 (1.3)        | 16.84 (1.1)        | 18.94 (0.9)        | 21.38 (1.3)        |
| $c^2$         | 24.57 (0.9)        | 19.97 (1.0)        | <b>20.07</b> (1.1) | 22.07 (1.4)        | 16.21 (1.0)        | 18.68 (0.9)        | 21.28 (1.1)        |
| $c^3$         | 28.07 (0.9)        | 20.40 (1.1)        | 20.72 (1.0)        | <b>21.61</b> (1.2) | 19.72 (1.1)        | 20.19 (0.8)        | 21.43 (1.1)        |
| $c^6$         | 34.52 (1.2)        | 21.28 (1.1)        | 21.28 (1.3)        | 24.14 (1.4)        | <b>13.56</b> (1.0) | 19.55 (0.9)        | 23.01 (1.3)        |
| $c^7$         | 28.84 (1.0)        | 20.00 (1.1)        | 20.18 (1.1)        | 22.44 (1.3)        | 15.60 (1.1)        | <b>18.57</b> (0.9) | 21.46 (1.3)        |
| $c^8$         | 27.93 (0.9)        | 19.85 (1.0)        | 20.10 (1.1)        | 21.78 (1.3)        | 17.48 (1.1)        | 18.96 (0.9)        | <b>21.13</b> (1.2) |

## 6 Agreement with Tables 2–5

Table 11 compares the 49 cells of each table with the paper, for the earlier one-margin-per-state model (“state”) and for the general PMC (“pair”).

Table 11: Difference with CSDA Tables 2–5 (our error minus the paper’s, in points) and cell-by-cell correlation. “state”:  $f_{ij} = f_{i1}$ ,  $s$  read as a standard deviation (earlier reproduction, 300 replications); “pair”: the four margins of Table 1,  $s$  read as a variance (30 replications).

| paper table | setting                    | mean $ \Delta $ (points) |             | mean $\Delta$ on the diagonal |       | correlation |       |
|-------------|----------------------------|--------------------------|-------------|-------------------------------|-------|-------------|-------|
|             |                            | state                    | pair        | state                         | pair  | state       | pair  |
| 2(a)        | PMC, $\tau = 0.16$ , Gauss | 6.98                     | <b>0.38</b> | +6.64                         | −0.29 | 0.876       | 0.988 |
| 2(b)        | PMC, $\tau = 0.16$ , Gamma | 12.04                    | <b>0.29</b> | +11.54                        | −0.30 | 0.877       | 0.991 |
| 3(a)        | PMC, $\tau = 0.70$ , Gauss | 8.24                     | <b>4.78</b> | +7.19                         | +4.67 | 0.884       | 0.924 |
| 3(b)        | PMC, $\tau = 0.70$ , Gamma | 12.31                    | <b>0.31</b> | +9.89                         | +0.11 | 0.794       | 0.999 |
| 4(a)        | PMM, $\tau = 0.16$ , Gauss | 7.74                     | <b>0.49</b> | +7.62                         | +0.50 | 0.901       | 0.993 |
| 4(b)        | PMM, $\tau = 0.16$ , Gamma | 11.17                    | <b>2.58</b> | +11.28                        | −2.43 | 0.889       | 0.914 |
| 5(a)        | PMM, $\tau = 0.70$ , Gauss | 7.10                     | <b>1.76</b> | +6.84                         | +1.05 | 0.539       | 0.749 |
| 5(b)        | PMM, $\tau = 0.70$ , Gamma | 18.29                    | <b>6.48</b> | +19.37                        | +6.49 | 0.660       | 0.697 |

- **The general PMC reproduces the paper.** Tables 2(a), 2(b), 3(b) and 4(a) are matched to within 0.3–0.5 point on average and a cell correlation of 0.988 or more. With one margin per state the errors were 7 to 12 points too high on average on the diagonal: the earlier reproduction simulated a different model.
- **Table 3(a) keeps a bias of +4.7 points** on the diagonal (PMC, Gaussian margins,  $\tau = 0.70$ ), although the same design with Gamma margins, Table 3(b), is reproduced to +0.1 point. The supervised error of the unsupervised experiment with Gaussian margins shows the same excess (Section 7.4). We could not identify the cause (see below).

- **Tables 4(b) and 5(b)** (PMM, Gamma margins) remain off by  $-2.4$  and  $+6.5$  points on the diagonal. The printed Table 5(b) is hard to reconcile with Table 5(a) and with Table 4(b): its errors with strong dependence (10.1–12.0 %) are well below its own product-copula column (17.0 %), whereas the paper’s Tables 4(a), 4(b) and 5(a) all show dependence *raising* the error of a pair-wise classifier.

**What the Table 3(a) gap is not.** The excess is confined to the PMC with Gaussian margins at  $\tau = 0.70$  and to copula-aware restorations: restoring the same data with the product copula (row  $c^0$ ) matches the paper to within 1–1.7 points, and the PMM of Table 5(a), which uses the same pair density (1) without the chain dynamics, matches to  $+1.1$  points on the diagonal. We tested seven explanations on the diagonal cells (20–30 replications), requiring each to fix Table 3(a) without breaking the tables that already agree:

- reading  $s$  as a standard deviation: Table 3(a)  $+3.3$  on average, Table 2(a) degraded to  $-1.1$ ;
- exchanging  $f_{12}$  and  $f_{21}$ , with either reading of  $s$ : Table 3(a)  $-0.8$  and  $-0.7$ , but Tables 2(a), 2(b) and 3(b) move by  $-2.7$  to  $-5.2$  points;
- simulating  $X$  as a Markov chain (Eq. 13 ignoring  $y_n$ ): Table 3(a)  $+9.1$ ;
- Eq. 13 evaluated with  $f_{ji}(y_n)$  instead of  $f_{ij}(y_n)$ : Table 3(a)  $-0.3$ , but Tables 2(a), 2(b) and 3(b) move to  $-2.5$ ,  $-2.7$  and  $-3.7$ , and the product-copula cell to 9.4 against 11.06 %;
- reading  $s$  as a variance in Eq. 13 and as a standard deviation in Eq. 14 and the copula arguments, or the reverse: Table 3(a)  $+3.7$  and  $+3.4$ .

Three independent implementations of the exact model — the package, a stand-alone simulator and a separately written forward–backward — give the same error rates. The gap is therefore not a coding error in the model as the paper states it; the 2013 computation of Table 3(a) presumably differed in a way the paper does not describe. It remains open.

## 7 Experiment 3 — Unsupervised copula selection by ICE (*CSDA Tables 6 and 7*)

### 7.1 Setup

$N = 2500$ ,  $K = 2$ ,  $Q = 30$  ICE iterations, margins known (their parameters are not re-estimated, whereas §4.3 of the paper re-estimates them). Two designs, each with the original prior  $p_{\text{orig}}$  and a balanced prior  $p_{\text{bal}}$  ( $p_{ii} = 0.35$ ,  $p_{12} = p_{21} = 0.15$ ):

- exp. 1: candidates  $\{c^1, c^3, c^6\}$  (Gauss, Gumbel, Clayton), true copulas  $c_{11} = c^1$  ( $\tau = 0.7$ ),  $c_{12} = c_{21} = c^3$  ( $\tau = 0.4$ ),  $c_{22} = c^6$  ( $\tau = 0.7$ ), Gaussian margins;
- exp. 2: candidates  $\{c^2, c^3, c^5, c^6\}$  (Student, Gumbel, cubic section, Clayton), true copulas  $c_{11} = c^2$  ( $\tau = 0.25$ ),  $c_{12} = c_{21} = c^5$  ( $\tau = 0.10$ ),  $c_{22} = c^6$  ( $\tau = 0.20$ ), Gamma margins.

At every M-step, ICE re-selects the family of each copula among the candidates and estimates its  $\tau$ . The package’s ICE estimates copulas by weighted pseudo-likelihood under the pair posteriors  $\xi_n(i, j)$ ; the paper (§4.2, Eqs. 22–24) estimates them on the hard sub-samples of a single posterior draw ( $L = 1$ ). The two coincide only when the posterior is concentrated. The paper selects the families with the Bayesian criterion of Huard, Évin and Favre [2] (its Eq. 20); we report that criterion here and compare six criteria in Section 8.

## 7.2 A single ICE start ends in the wrong basin

The first reproduction started ICE from every copula on the first candidate with  $\tau = 0$ . With Gaussian margins it rarely recovers the diagonal copulas:  $c_{11}$  in 2 % of runs with  $p_{\text{orig}}$ ,  $c_{11}$  and  $c_{22}$  in 0 % with  $p_{\text{bal}}$  (Table 12). Three checks locate the cause in the convergence of ICE rather than in the selection criterion:

1. *With the true parameters*, the M-step (MLE criterion) selects the true diagonal families in 10 seeds out of 10 in all four settings, whether it is fed the true labels, the posterior  $\xi$  or one posterior draw.
2. *Started at the true model*, ICE (MLE criterion, exp. 1) keeps the true families (diagonal 10/10) and ends with a log-likelihood higher than the single start by a median of 30 nat ( $p_{\text{orig}}$ ) and 58 nat ( $p_{\text{bal}}$ ) over 10 seeds. Its unsupervised error with  $p_{\text{orig}}$  is 13.7 %, the paper’s 13.74 %.
3. *The k-means warm start of the package* was no remedy: it numbered the states arbitrarily and, with known margins, estimated the prior and copulas on permuted labels in about half the starts. This defect is fixed (clusters are now matched to the declared state laws).

The family of a start decides the basin. The package’s own multistart, which only perturbs the parameters of a start, keeps its families and cannot leave that basin.

## 7.3 Multistart ICE

The multistart runs ICE from the single start above, from the corrected k-means start, and from every combination of candidate families on the two diagonal copulas ( $\tau = 0.5$ ; off-diagonal copulas on the first candidate,  $\tau = 0.3$ ), and keeps the run of highest final log-likelihood. No start uses the truth. Table 12 compares it with the single start and with the paper.

Table 12: Right copula family chosen (%) and mean error rate (%), Huard criterion, pair margins. Paper: Tables 6–7 (10 runs) and the error rates of its §4.3 (given for  $p_{\text{orig}}$  only).

| setting                   | source                  | right family chosen (%) |          |          |          | error (%) |       |
|---------------------------|-------------------------|-------------------------|----------|----------|----------|-----------|-------|
|                           |                         | $c_{11}$                | $c_{12}$ | $c_{21}$ | $c_{22}$ | unsup.    | sup.  |
| exp. 1, $p_{\text{orig}}$ | paper (10 runs)         | 100                     | 90       | 70       | 100      | 13.74     | 12.16 |
|                           | single start (100 runs) | 2                       | 79       | 91       | 97       | 17.9      | 14.4  |
|                           | multistart (30 runs)    | 100                     | 73       | 83       | 100      | 14.7      | 14.4  |
| exp. 2, $p_{\text{orig}}$ | paper (10 runs)         | 100                     | 90       | 90       | 100      | 12.89     | 11.35 |
|                           | single start (100 runs) | 100                     | 64       | 56       | 97       | 10.7      | 10.5  |
|                           | multistart (30 runs)    | 100                     | 70       | 43       | 97       | 10.3      | 10.2  |
| exp. 1, $p_{\text{bal}}$  | paper (10 runs)         | 100                     | 100      | 100      | 100      | –         | –     |
|                           | single start (100 runs) | 0                       | 53       | 59       | 0        | 31.0      | 20.2  |
|                           | multistart (30 runs)    | 100                     | 60       | 47       | 100      | 20.9      | 20.4  |
| exp. 2, $p_{\text{bal}}$  | paper (10 runs)         | 100                     | 100      | 100      | 80       | –         | –     |
|                           | single start (100 runs) | 100                     | 65       | 62       | 77       | 20.9      | 20.7  |
|                           | multistart (30 runs)    | 97                      | 80       | 60       | 97       | 20.7      | 20.4  |

- **Diagonal copulas are recovered** in 97–100 % of runs in all four settings (paper: 80–100 %). With Gaussian margins the winning start is a diagonal sweep in every run (Huard criterion).
- **The unsupervised error comes within 0.5 point of the supervised one** (14.7 vs 14.4 %, 10.3 vs 10.2 %, 20.9 vs 20.4 %, 20.7 vs 20.4 %): estimation costs almost nothing once ICE starts in the right basin. From a single start, the loss with Gaussian margins was 3.4 and 10.8 points.

- **Off-diagonal copulas stay mostly below the paper:** 43–83 % against 70–100 % (only  $c_{21}$  of exp. 1,  $p_{\text{orig}}$ , exceeds the paper’s 7 out of 10). These blocks receive about 125 ( $p_{\text{orig}}$ ) to 375 ( $p_{\text{bal}}$ ) effective observations. Even with the true labels and parameters, the cubic-section copula of exp. 2,  $p_{\text{orig}}$ , is selected in only 6 seeds out of 10 (MLE criterion). In a pilot study on exp. 2,  $p_{\text{orig}}$ , of the earlier one-margin-per-state benchmark, weighted Vuong tests [3] declared these choices statistically tied with the runner-up in 75–100 % of the runs, the true family always belonging to the set of families not significantly worse than the selected one. The paper’s 9–10 out of 10 rest on ten runs.

## 7.4 What remains

With Gaussian margins and  $p_{\text{orig}}$ , the *supervised* error of our simulations is 14.4 % against 12.16 % in the paper. This excess does not involve ICE and has the same sign as the bias of Table 3(a); with Gamma margins our supervised error is lower than the paper’s (10.2 vs 11.35 %). Neither is explained by the hypotheses of Section 6.

# 8 Experiment 4 — Comparison of selection criteria

## 8.1 Criteria

**MLE**  $\arg \max_r \log L_r(\hat{\tau}_r)$ , the weighted log-likelihood at the fitted parameter.

**AIC**  $\arg \max_r 2 \log L_r - 2k_r$ .

**BIC**  $\arg \max_r 2 \log L_r - k_r \log n_{\text{eff}}$ , with  $n_{\text{eff}} = \sum_n \xi_n(i, j)$ .

**Huard**  $\arg \max_r \frac{1}{\tau_M^r - \tau_m^r} \int_{\tau_m^r}^{\tau_M^r} \prod_n c_r(u_n, v_n; \tau)^{\xi_n} d\tau$  (paper, Eq. 20): Bayesian evidence under a uniform prior on each family’s admissible  $\tau$ -range, computed by adaptive quadrature.

**CvM**  $\arg \min_r S_n(c_r)$ , the weighted Cramér–von Mises distance between empirical and fitted copula CDFs.

**xv-CIC** cross-validated copula information criterion: out-of-sample log-likelihood on contiguous, purged folds.

## 8.2 With pair margins and multistart

Table 13 summarises the six criteria on the reproduction of Section 7 ( $D$ : diagonal recovery,  $O$ : off-diagonal recovery,  $\bar{e}$ : unsupervised error; 30 runs).

Table 13: Selection-criterion comparison, pair margins and multistart ICE (30 runs).  $D$  /  $O$  = diagonal / off-diagonal recovery (%);  $\bar{e}$  = mean unsupervised classification error (%).

| config                    | MLE        |    |             | AIC        |    |           | BIC        |    |             | Huard      |           |             | CvM |           |           | xv-CIC     |    |           |
|---------------------------|------------|----|-------------|------------|----|-----------|------------|----|-------------|------------|-----------|-------------|-----|-----------|-----------|------------|----|-----------|
|                           | D          | O  | $\bar{e}$   | D          | O  | $\bar{e}$ | D          | O  | $\bar{e}$   | D          | O         | $\bar{e}$   | D   | O         | $\bar{e}$ | D          | O  | $\bar{e}$ |
| exp. 1, $p_{\text{orig}}$ | <b>100</b> | 53 | 14.7        | <b>100</b> | 53 | 14.7      | <b>100</b> | 53 | 14.7        | <b>100</b> | <b>78</b> | <b>14.7</b> | 70  | 55        | 18.3      | <b>100</b> | 60 | 14.7      |
| exp. 2, $p_{\text{orig}}$ | 97         | 35 | 10.4        | <b>98</b>  | 45 | 10.3      | 97         | 52 | <b>10.3</b> | <b>98</b>  | 57        | 10.3        | 35  | <b>60</b> | 10.5      | <b>98</b>  | 37 | 10.3      |
| exp. 1, $p_{\text{bal}}$  | <b>100</b> | 45 | 20.9        | <b>100</b> | 45 | 20.9      | <b>100</b> | 45 | 20.9        | <b>100</b> | 53        | <b>20.9</b> | 85  | <b>55</b> | 21.6      | <b>100</b> | 45 | 20.9      |
| exp. 2, $p_{\text{bal}}$  | <b>98</b>  | 53 | <b>20.6</b> | <b>98</b>  | 55 | 20.7      | <b>98</b>  | 52 | 20.7        | 97         | <b>70</b> | 20.7        | 40  | 62        | 20.9      | <b>98</b>  | 40 | 20.7      |

$D$  = diagonal recovery rate (%; higher is better,  $K=2$  diagonal cells  $\times$  runs);  $O$  = off-diagonal recovery rate (%);  $\bar{e}$  = mean unsupervised classification error (%; lower is better). Bold = best per row.

MLE, AIC, BIC, Huard and xv-CIC all recover the diagonal copulas in 97–100 % of runs and reach the same error to within 0.1 point. They differ off-diagonal only, where Huard is the best likelihood-based criterion in every setting (78, 57, 53 and 70 % against 53, 35, 45 and 53 % for MLE). CvM is the only criterion that fails on the diagonal (35–85 %) and it has the highest error with Gaussian margins.

### 8.3 The earlier benchmark: one margin per state, single start

The rest of this section analyses the earlier design — one margin per state and a single ICE start, 100 runs — in which the criteria were first compared and in which the prior-width effect behind Huard’s off-diagonal advantage was isolated. That design is not the paper’s model and its error rates are inflated by the wrong ICE basin, but it is a controlled benchmark for comparing criteria on identical data, and its conclusions on the criteria stand. Its summary is Table 14; each rate pools 200 Bernoulli trials and the runs are paired across criteria (run  $r$  uses seed  $3000 + r$ ), so comparisons use McNemar’s exact test on per-run outcomes.

Table 14: Selection-criterion comparison (summary). D / O = diagonal / off-diagonal recovery (%);  $\bar{e}$  = mean unsupervised classification error (%).

| config                    | MLE |    |           | AIC |    |           | BIC       |    |             | Huard |           |           | CvM       |           |             | xv-CIC    |    |           |
|---------------------------|-----|----|-----------|-----|----|-----------|-----------|----|-------------|-------|-----------|-----------|-----------|-----------|-------------|-----------|----|-----------|
|                           | D   | O  | $\bar{e}$ | D   | O  | $\bar{e}$ | D         | O  | $\bar{e}$   | D     | O         | $\bar{e}$ | D         | O         | $\bar{e}$   | D         | O  | $\bar{e}$ |
| exp. 1, $p_{\text{orig}}$ | 57  | 27 | 28.0      | 57  | 27 | 28.0      | 57        | 27 | 28.0        | 56    | <b>35</b> | 28.3      | 42        | 27        | <b>27.1</b> | <b>58</b> | 31 | 27.8      |
| exp. 2, $p_{\text{orig}}$ | 97  | 7  | 22.0      | 98  | 6  | 22.0      | <b>98</b> | 6  | <b>22.0</b> | 98    | <b>41</b> | 22.0      | 14        | 38        | 24.3        | 97        | 4  | 22.0      |
| exp. 1, $p_{\text{bal}}$  | 50  | 12 | 31.7      | 50  | 12 | 31.7      | 50        | 12 | 31.7        | 50    | 18        | 31.9      | <b>62</b> | <b>48</b> | <b>28.8</b> | 50        | 14 | 31.7      |
| exp. 2, $p_{\text{bal}}$  | 90  | 8  | 32.6      | 94  | 16 | 32.5      | <b>98</b> | 6  | <b>32.4</b> | 91    | 38        | 32.5      | 17        | <b>41</b> | 33.5        | 94        | 9  | 32.6      |

$D$  = diagonal recovery rate (%; higher is better,  $K=2$  diagonal cells  $\times$  runs);  $O$  = off-diagonal recovery rate (%);  $\bar{e}$  = mean unsupervised classification error (%; lower is better). Bold = best per row.

**(1) The likelihood-based criteria agree with one another.** MLE, AIC and BIC return identical recovery rates on both exp. 1 settings and are within 1.5 points on exp. 2,  $p_{\text{orig}}$ . They part company on exp. 2,  $p_{\text{bal}}$ , where AIC reaches  $O = 16\%$  against MLE’s  $8\%$  (McNemar 20:5,  $p = 0.004$ ) and BIC  $D = 98\%$  against  $90\%$  ( $16:0$ ,  $p = 3.1 \times 10^{-5}$ ). At these sample sizes the penalties only arbitrate near-ties. Every one of them recovers the two-parameter Student copula of exp. 2 in  $100\%$  of runs.

**(2) Huard dominates off-diagonal.** Huard matches MLE on the diagonal ( $56.5/98/50/91\%$  against  $57/97/50/90\%$ ), although BIC and AIC do better there on exp. 2,  $p_{\text{bal}}$  ( $98$  and  $94.5\%$ ; BIC against Huard  $14:0$ ,  $p = 1.2 \times 10^{-4}$ ). Off-diagonal, Huard is higher than MLE in all four settings:  $35$  vs  $27\%$  (exp. 1,  $p_{\text{orig}}$ ),  $41$  vs  $7\%$  (exp. 2,  $p_{\text{orig}}$ ),  $17.5$  vs  $11.5\%$  (exp. 1,  $p_{\text{bal}}$ ) and  $38.5$  vs  $8\%$  (exp. 2,  $p_{\text{bal}}$ ). All four gaps are significant against MLE:  $19:3$  ( $p = 8.6 \times 10^{-4}$ ),  $68:0$  ( $p = 6.8 \times 10^{-21}$ ),  $12:0$  ( $p = 4.9 \times 10^{-4}$ ) and  $61:0$  ( $p = 8.7 \times 10^{-19}$ ).

Sparsity alone does not explain it. Both experiments share the off-diagonal sparsity of  $p_{\text{orig}}$ , but the size of Huard’s edge depends on the true family (Table 15):  $+4$  to  $+12$  points when it is Gumbel, whose  $\tau$ -range  $[0, 1]$  has width 1, and  $+30$  to  $+36$  points when it is the cubic-section copula, whose range  $[0, 0.165]$  is six times narrower. Eq. (20) places a uniform prior on each family’s own range, so a narrow range concentrates prior mass.

Table 15: Off-diagonal recovery by cell, Huard vs MLE (earlier benchmark), and McNemar discordance (Huard only : MLE only).

| setting                   | true family (width) | Huard ( $c_{12}$ , $c_{21}$ ) | MLE ( $c_{12}$ , $c_{21}$ ) | discordance |
|---------------------------|---------------------|-------------------------------|-----------------------------|-------------|
| exp. 1, $p_{\text{orig}}$ | GH (1.000)          | 26%, 44%                      | 22%, 32%                    | 6:2, 13:1   |
| exp. 2, $p_{\text{orig}}$ | CubSec (0.165)      | 42%, 40%                      | 10%, 4%                     | 32:0, 36:0  |
| exp. 1, $p_{\text{bal}}$  | GH (1.000)          | 20%, 15%                      | 12%, 11%                    | 8:0, 4:0    |
| exp. 2, $p_{\text{bal}}$  | CubSec (0.165)      | 36%, 41%                      | 6%, 10%                     | 30:0, 31:0  |

*An isolation experiment settles it.* The `huard_common` criterion is the same evidence integrated over one *shared* support — the intersection of the candidates’ declared ranges — so that no family is rewarded for declaring a narrow one. Same data, truth, candidates and seeds; only the prior support changes (Table 16).

Table 16: Off-diagonal recovery with Huard’s own supports and with a common support. “Edge retained”: share of Huard’s advantage over MLE that survives the common support. Discordance: Huard only : common only.

| setting                   | true family | MLE   | Huard | common support | edge retained | Huard vs common | $p$                   |
|---------------------------|-------------|-------|-------|----------------|---------------|-----------------|-----------------------|
| exp. 1, $p_{\text{orig}}$ | GH          | 27.0% | 35.0% | 30.0%          | 38%           | 17:7            | 0.06                  |
| exp. 2, $p_{\text{orig}}$ | CubSec      | 7.0%  | 41.0% | 8.0%           | 3%            | 66:0            | $2.7 \times 10^{-20}$ |
| exp. 1, $p_{\text{bal}}$  | GH          | 11.5% | 17.5% | 11.5%          | 0%            | 12:0            | $4.9 \times 10^{-4}$  |
| exp. 2, $p_{\text{bal}}$  | CubSec      | 8.0%  | 38.5% | 7.0%           | -3%           | 63:0            | $2.2 \times 10^{-19}$ |

With a common support, Huard’s advantage disappears in three settings (a retained share of 3, 0 and  $-3\%$ ) and  $38\%$  of it —  $+3.0$  points,  $p = 0.03$  — survives in the fourth. **Huard’s off-diagonal superiority in this benchmark is almost entirely an effect of the families’ declared  $\tau$ -ranges, not of averaging the likelihood over  $\tau$ .** It is genuine Bayesian behaviour — a narrower prior that contains the truth *ought* to win — but it is a property of the candidate set, and it will not transfer to candidate sets whose ranges are comparable. A methodological note developing this point is in preparation.

**(3) The prior of the chain matters more than the criterion.** Within a setting, the unsupervised errors of the six criteria span about 1 to 3 points; between settings they range from  $22.0\%$  (exp. 2,  $p_{\text{orig}}$ , BIC) to  $33.5\%$  (exp. 2,  $p_{\text{bal}}$ , CvM). Going from  $p_{\text{orig}}$  to  $p_{\text{bal}}$  raises the error by 1.6 to 10.6 points depending on the criterion and the design: balanced priors switch states more often and weaken the Markov contrast the classifier relies on.

**(4) Cramér–von Mises trades diagonal for off-diagonal.** CvM has the lowest error on both exp. 1 settings ( $27.1\%$  and  $28.8\%$ ) and the highest off-diagonal recovery of the benchmark ( $O = 48\%$  on exp. 1,  $p_{\text{bal}}$ , beating Huard by 74 cells to 13,  $p = 1.6 \times 10^{-11}$ ), but its diagonal recovery collapses to 14 and  $17\%$  on exp. 2, where it almost never selects the Student copula. The  $L^2$  distance weights every deviation equally, including where the density is small: a liability when two families differ in the tails, an asset on sparse blocks where a likelihood overfits. The multistart reproduction confirms its diagonal weakness (Table 13).

**(5) xv-CIC: better founded, not better performing.** It tracks AIC and BIC on the diagonal ( $58/97/50/94\%$ ) and lies between MLE and Huard off-diagonal in three settings, but it is the worst of the six off-diagonal on exp. 2,  $p_{\text{orig}}$  ( $O = 4\%$ ), where Huard beats it by 74 cells to 0 ( $p = 1.1 \times 10^{-22}$ ). Cross-validation lets the effective penalty adapt to the family, which suits plug-in margins and posterior weights better than a fixed penalty; at these sample sizes the fixed penalty is good enough, and xv-CIC cost about 5.6 times an MLE evaluation per block on a synthetic benchmark.

**On statistical power.** The conclusions on Huard rest on discordances as one-sided as 68:0, 61:0 and 12:0. Two comparisons are inconclusive and reported as such: Huard vs CvM off-diagonal on exp. 2 ( $p = 0.52$  and  $p = 0.66$ ) and Huard vs xv-CIC on exp. 1,  $p_{\text{orig}}$  ( $p = 0.22$ ).

#### Practical recommendation.

- **Start ICE several times**, varying the copula families of the starts; the criterion matters far less than the basin.
- **For copula identification, Huard** is as good as MLE on the diagonal and better off-diagonal, but its edge comes from the declared  $\tau$ -ranges. Use it when a candidate’s plausible range is genuinely narrow; do not expect it when the ranges are comparable.

- **MLE, AIC and BIC** are sound defaults when the goal is classification; they agree except on near-ties.
- **CvM** is a specialist for suspect likelihoods or rare transitions, never a default: it cannot identify the Student copula.
- **xv-CIC** is a diagnostic rather than a default.
- Off-diagonal choices on a hundred observations are often statistical ties; test them (Vuong) before interpreting them.

**Revision notes.** (a) An earlier version of this section was computed with the weighted log-likelihood returned as a per-observation mean while the AIC and BIC penalties are calibrated on a sum; the penalty was  $n_{\text{eff}}$  times too strong and the Huard integral was flattened. Its conclusions (AIC/BIC excluding the Student copula, Huard trading diagonal for off-diagonal recovery) did not survive the correction. (b) The numbers of the earlier benchmark were regenerated after the numerical corrections of the copula kernels and of the Huard quadrature; 27 of the 9600 per-cell decisions changed. (c) The present version adds the general PMC, the variance reading of Table 1, the paper’s table orientation and the multistart ICE.

## 9 Conclusion

1. **The model matters first.** With pair-indexed margins — the general PMC of the paper — and the variance reading of Table 1, four of the eight supervised tables are reproduced to within half a point on average; with one margin per state, which is a hidden Markov chain, every table was 7 to 18 points off on average.
2. **The qualitative conclusions of the paper hold.** The true copula is the best restoration copula (27 columns out of 28); the copula shape matters at  $\tau = 0.70$ , where a wrong copula can multiply the error by 3.5 to 5.7; Markovianity lowers the error by 32–42 % at  $\tau = 0.16$ .
3. **ICE with automatic copula selection works once it starts in the right basin.** A multistart over the diagonal copula families recovers the diagonal copulas in 97–100 % of runs and loses at most about half a point of error with respect to the supervised restoration. Off-diagonal copulas estimated on about a hundred observations are often statistically undecidable.
4. **Open gaps:** Table 3(a) and the supervised error with Gaussian margins (about +2 to +5 points), unexplained by seven tested hypotheses on the margins and the simulator, and the PMM Tables 4(b) and 5(b).

## A Copula registry

## B Result files

- `report/results/margins_pair/` — Experiments 1–3, pair margins, single ICE start (the tables of Sections 4, 5 and the single-start rows of Table 12);
- `report/results/exp3_multistart/margins_pair/` — Experiment 3 with multistart ICE;
- `report/results/` — the earlier one-margin-per-state results (Section 8, earlier benchmark) and the prior-width isolation (`prior_width_isolation*.csv`);
- `report/tables/` — the corresponding L<sup>A</sup>T<sub>E</sub>X tables; `report/tables/synthesis/` — the comparison tables of this report.

Compile with `cd report && latexmk -pdf csda2013_reproduction.tex`.

Table 17: Copulas of this report, matching CSDA Table A.11. The Student copula has  $\nu = 4$  in Experiments 1–2; in Experiments 3–4 ICE estimates  $\nu$  as well.

| Code  | Family             | Package class               | Admissible $\tau$ |
|-------|--------------------|-----------------------------|-------------------|
| $c^0$ | Product            | <code>CopulaProduct</code>  | $\{0\}$           |
| $c^1$ | Gauss (elliptical) | <code>CopulaGaussian</code> | $[-1, 1]$         |
| $c^2$ | Student-t          | <code>CopulaStudent</code>  | $[-1, 1]$         |
| $c^3$ | Gumbel–Hougaard    | <code>CopulaGH</code>       | $[0, 1]$          |
| $c^4$ | FGM                | <code>CopulaFGM</code>      | $[-2/9, 2/9]$     |
| $c^5$ | Cubic section      | <code>CopulaCubSec</code>   | $[0, 33/200]$     |
| $c^6$ | Clayton            | <code>CopulaClayton</code>  | $(0, 1]$          |
| $c^7$ | Archimedean 12     | <code>CopulaA12</code>      | $[1/3, 1)$        |
| $c^8$ | Archimedean 14     | <code>CopulaA14</code>      | $[1/3, 1)$        |

## References

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